

A perfect shuffle has a clock

Name: _____

Date: _____

Use eight cards 0–7 in a row, leftmost = top. Cut into two equal halves **without reversing either half**. Interleave, starting with the first half: 0 1 2 3 / 4 5 6 7 becomes **0 4 1 5 2 6 3 7**. This is an out-shuffle.

1. Shuffle until home.

Repeat the exact rule. How many shuffles first restore the whole original order? Record rows or track each card's *position*.

My rows or position cycles:

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2. Explain the rule with remainders.

Positions are numbered 0–7. Track where a card at old position x goes. For $x=0$ through 6, test “double x , then take the remainder after division by 7.” What happens to position 7?

Old position	0	1	2	3	4	5	6	7
New position								

3. Prove the return time.

After t shuffles, a position below 7 becomes the remainder of $2^t x$ after division by 7. Find the first t with 2^t leaving remainder 1. Why does checking $x=1$ show that no earlier return works?

My proof, including the fixed last card:

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Go further: Predict the first return for 6 cards and for 16 cards. For $2m$ cards the relevant modulus is $2m - 1$. Why does twice as many cards not necessarily mean twice as many shuffles? **Research bridge:** Diaconis, Graham, and Kantor studied which permutations combinations of perfect in- and out-shuffles generate. This page proves the single out-shuffle return formula in a small case.

Try it with objects. Explain by pointing, talking, or drawing.